# BA0-R systems-modeling prior-art synthesis

Revision 3 - research history

Baseline: `89e5486e02d07ff5b97082f73a2e22ac1b4319ae`

**Status**

Includes SRC-0048, the full-text ISO/IEC/IEEE 42010:2011 regression. Semantic closure remains OPEN; no BAE metaclass is promoted.

## 1 Corpus and access boundary

Revision 3 preserves the R2 corpus and adds SRC-0048, a user-supplied full-text reading of ISO/IEC/IEEE 42010:2011. SRC-0043 remains the separate ACCESS-LIMITED 2022 record; no equivalence between editions is assumed.

## 2 Material correction from ISO 42010:2011

The earlier direction `canonical model -> projection -> view` was too strong as a statement of general prior art. ISO 42010 describes both synthetic construction of views integrated through correspondences and projective extraction of views from an underlying repository.

```
one canonical repository is universally mandatory   REJECTED as prior-art claim
projective DDTA Base Analysis                       CANDIDATE design choice
```

DDTA may still choose one accepted Base Analysis to simplify consistency, provenance, refresh and re-analysis, but BA0/BA4 must justify that choice.

## 3 View terminology

| | |
|---|---|
| **Viewpoint** | reusable conventions framing concerns and specifying model kinds plus construction/interpretation/analysis methods. |
| **View** | application of one viewpoint to one system-of-interest; its architecture models collectively address the framed concerns across the whole system. |
| **Architecture Model** | model governed by a model kind; may selectively present part of the system and may be shared by multiple views. |

A partial diagram or filtered graph should not automatically be called an architecture view if DDTA adopts ISO-aligned terminology.

## 4 ISO 42010 does not solve BA1

The standard explicitly does not define what a system is. Its ontology governs architecture-description work products, not the internal system-model ontology. Annex B nevertheless provides a useful metamodel lens: entities, attributes, relationships and constraints.

## 5 Cross-source responsibility status

| Responsibility | Evidence after NASA/SysML/KerML/AADL/OPM/ISO | DDTA status |
|---|---|---|
| Structure / referent | strong across traditions | semantic responsibility STRONG; exact taxonomy OPEN |
| Behavior / transformation | function/action/behavior/Process evidence | STRONGLY EVIDENCED |
| Information/resource | recurring but represented differently | semantic need strong; universal InformationObject root OPEN/-WEAKENED |
| Relation/connection | connector/link/correspondence evidence | typed relation semantics STRONGLY EVIDENCED |
| Interface/endpoint | recurrent, often viewpoint/model-kind specific | first-class identity OPEN |
| Flow/transfer | recurrent; can be relation semantics | root OPEN/WEAKENED |
| State/mode | recurrent; can be owned/classified | root OPEN/WEAKENED |
| Boundary/environment | distinction strong, root not forced | scope distinction STRONG; Boundary root OPEN/WEAKENED |
| Property/constraint | explicit across metamodels | likely property/constraint mechanism |
| View/viewpoint/model kind | direct ISO full-text support | distinctions STRONGLY EVIDENCED |
| Cross-model relations | ISO correspondences plus other approaches | STRONGLY EVIDENCED prior art |
| Analysis | SysML/AADL/NASA/ISO | generic model-based analysis is NOT DDTA novelty |

## 6 Hypothesis changes

Actor root and Asset root remain REJECTED as generic roots. Interaction remains REJECTED as a sufficient universal relation. Channel, Store, Contract, Boundary, State, Flow and InformationObject remain OPEN/-WEAKENED as mandatory roots. ISO additionally rejects the claim that a single repository is required by architecture-description prior art.

## 7 Correspondence and provenance pressure

ISO correspondences can be explicit n-ary relations governed by rules and used for consistency, traceability, composition, refinement, dependency and model transformation, including links between architecture descriptions and requirements. DDTA historical analysis-to-governed-change provenance remains a distinct unresolved problem, but BA3/BA4 must compare against correspondence prior art.

## 8 Decision and rationale regression

Architecture decisions and rationale, including affected elements, alternatives, consequences, timestamps and citations, are established prior art. DDTA Decision remains distinct because its closed responsibility is normative ownership/decomposition under one MacroRequirement. No Chapter 4 reopen trigger is produced.

## 9 Novelty challenge

Established prior art now includes modeling metamodels, multiple views/viewpoints, model-based analysis, cross-model consistency/traceability and decision rationale. The candidate DDTA contribution must remain the governed composition boundary:

```
governed documentation
 -> reviewed methodology-neutral analyzable representation
 -> replaceable method-owned security analysis
 -> reviewed normalized findings
 -> accepted governed document / requirement change
 -> preserved analysis-to-change provenance
 -> change-aware re-analysis
```

The reviewed systems-modeling corpus does not establish that entire loop. This is a bounded comparative inference, not proof of novelty.

## 10 Chapter regression

```
Chapter 3 reopen trigger  NOT FOUND
Chapter 4 reopen trigger  NOT FOUND
```

## 11 BA0 working hypothesis - OPEN

**Working hypothesis**

Base Analysis is an accepted, methodology-neutral analyzable representation of governed project/system knowledge, grounded in governed documentation and explicit reviewed analytical additions. It must preserve analysis-relevant semantic distinctions without adopting one external modeling-language ontology. DDTA may choose a canonical projective representation to support repeatable views, consistency, provenance and re-analysis, but that choice must be justified against synthetic multi-view alternatives rather than assumed as a universal modeling principle.

## 12 Next decision

After the history package is committed, decide whether to merge the candidate sources into the canonical literature registry and close BA0-R. No BAE metaclass is accepted here.